AHNAF ATIQUE

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Dhaka, Bangladesh

Objective

Final-year Computer Science and Engineering undergraduate with a strong interest in teaching and research. My experience as an Undergraduate Teaching Assistant in six core courses has deepened my commitment to helping students learn. My research interests lie in Machine Learning, Computer Vision and Computational Biology. I seek to learn from experienced faculty, contribute to meaningful research, and grow as both a teacher and a researcher.

Teaching Experience

- **United International University (UIU) [🌐]** *Nov 2024 - Oct 2026*
  Undergraduate Teaching Assistant (Part-time) Dhaka, Bangladesh
  - Assisted faculty in conducting lab sessions and tutorials for the following undergraduate courses:
    - Introduction to Computer Systems
    - Structured Programming Language
    - Object Oriented Programming
    - Data Structures and Algorithms
    - Microprocessors and Microcontrollers
    - Database Management Systems
  - Selected as invigilator for the Freshers' Coding Challenge 2025

Education

- **United International University (UIU)** *2026*
  B.Sc. in Computer Science and Engineering (Final Year) Dhaka, Bangladesh
  - CGPA: 3.94
- **Comilla Victoria Govt. College** *2021*
  Higher Secondary Certificate (HSC) Comilla, Bangladesh
  - GPA: 5.00
- **Comilla Zilla School** *2019*
  Secondary School Certificate (SSC) Comilla, Bangladesh
  - GPA: 5.00

Undergraduate Thesis (FYDP)

- **Enhancing Enhancer-Promoter Interaction Prediction via Gene Regulatory Inference and Hybrid Deep Learning Architectures (SLIM) [GitHub]**
  Supervisor: Dr. Riasat Azim, Associate Professor, Dept. of CSE, UIU *Sep 2026*
  - Built a memory-based deep learning model for gene-regulation prediction and a leakage-free evaluation protocol
  - Matched the performance of a published baseline while running 2.3× faster and using 50% less GPU memory

Research Interests

Machine Learning · Computer Vision · Computational Biology

Research

[1] **SLIM: Selective Memory and Leakage-Controlled Evaluation for Enhancer-Promoter Interaction Prediction**. *Ready for submission.*

[2] **Holo-GeoN: Structure-Free, Physics-Constrained Prediction of Mutation Effects on Protein Stability and Pathogenicity**. *Ready for submission.*

[3] **Bounded Waiting Time in Sensor-Driven Waste Collection: A Reserved-Head Dispatch Rule for Prize-Collecting Vehicle Routing**. *Ready for submission.*

[4] **Entity Leakage Overstates Measured Fraud-Detection Accuracy on IEEE-CIS**. *Manuscript in preparation.*

PROJECTS

- **Radflow-EDGE: Offline Chest X-ray Triage**
  Medical Imaging, Edge AI [GitHub]
  ◦ Built an offline, edge-native AI system that enables a single rural health worker to perform chest X-ray triage without cloud access

- **AI-CLI-Orchestrator: Multi-Agent AI Orchestration**
  Multi-Agent Systems, Large Language Models [GitHub]
  ◦ Developed a production-oriented system in which a central orchestrator (Grok, Gemini, DeepSeek) plans tasks and routes them to parallel CLI coding agents

- **NOVIS: Non-Optical Visual Inference System**
  Computer Vision, IoT, Sensor Fusion [GitHub]
  ◦ Developed training and inference code to reconstruct grayscale images, metric depth and inferred color from a 32×24 thermopile array, ultrasonic ranges and active chirp echoes, with no camera at inference time
  ◦ Built a custom data-collection device with thermal sensors, sonar sensors and a speaker-microphone pair

- **DiaFact: Diabetes Factor Analysis**
  Machine Learning, Explainable AI [GitHub]
  ◦ Built a machine learning pipeline for diabetes factor analysis with explainable AI techniques

- **Personalized Diet Recommendation System**
  Web Application [GitHub]
  ◦ Developed a web application that offers tailored dietary guidance, personalized meal plans and real-time calorie tracking, and connects users with professional dietitians

- **EduSync: Campus Solution Platform**
  Web Application [GitHub]
  ◦ Developed a one-stop platform for streamlined campus life management

- **PillTrack: Medication Tracking System**
  Health Application [GitHub]
  ◦ Built a medication tracking application to help users manage prescriptions

- **Mr.Bin: Smart Waste Management System**
  IoT, Embedded Systems [GitHub]
  ◦ Designed and built an IoT-based smart waste bin using embedded systems

SKILLS

- **Programming Languages:** Python, C, C++, Java, JavaScript, SQL
- **Machine Learning & Vision:** PyTorch, TensorFlow, Scikit-learn, OpenCV, Hugging Face, SHAP, NumPy, Pandas
- **Embedded Systems & Edge AI:** Raspberry Pi, ESP32, Arduino, Proteus, Logisim, Cisco Packet Tracer

HONORS AND AWARDS

- **Merit-Based Scholarship** *Every Trimester*
  United International University (UIU)
  ◦ Received merit-based scholarship in every trimester: 100% tuition waiver (7 times), 50% (2 times), 25% (3 times)

- **Microprocessor Project Show – Prize Winner**
  United International University (UIU)
  ◦ Awarded prize for innovative hardware project at the Microprocessor Project Show

REFERENCES

Dr. Riasat Azim
Associate Professor, Department of CSE
United International University, Dhaka
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